

# Distance Learning for Analog Methods

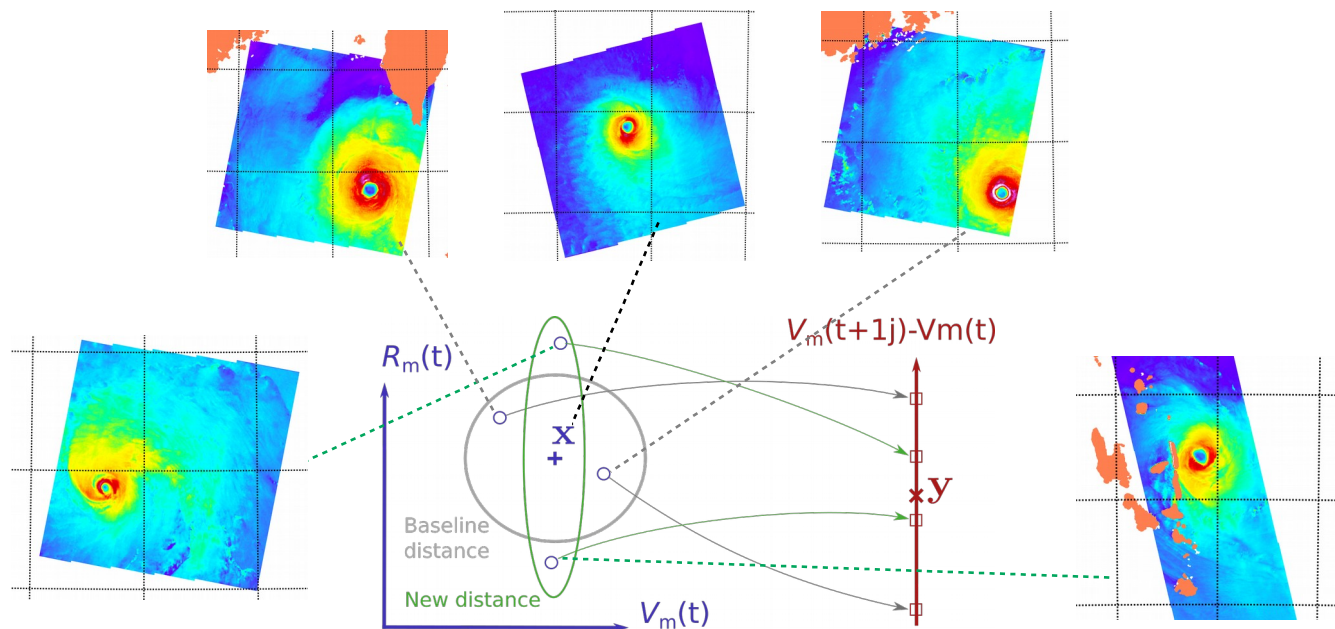
## Application to Tropical Cyclone Intensity Forecasting

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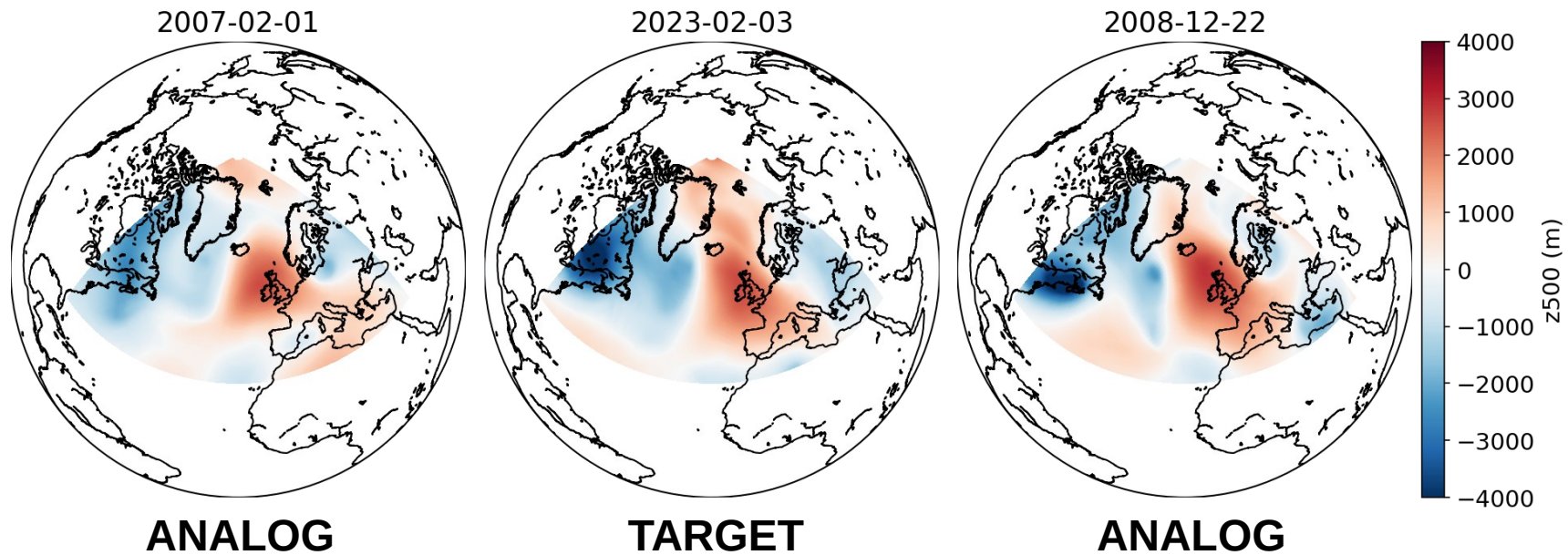
Lucas Drumetz, Alexis Mouche, Pierre Tandeo, Léo Vinour

LOPS-SIAM - Oceanix



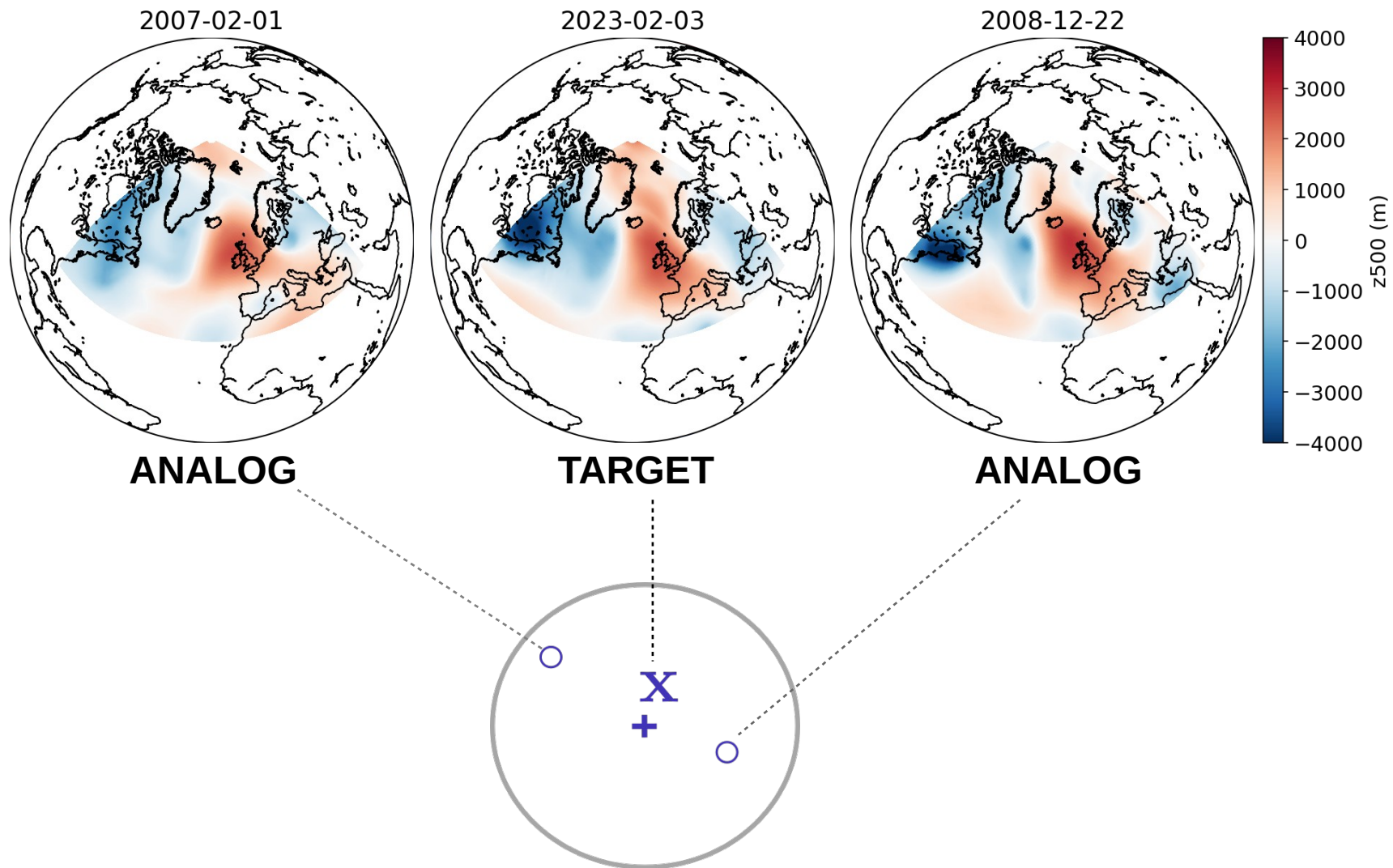
# Analog methods

- First paper : Lorenz (1969) → predictability of atmospheric flows
- Idea : « history repeats itself ».
- Example : ERA5 reanalysis, 500mb geopotential height anomaly



- Applications : Extreme event attribution (and others...)

# Analog methods



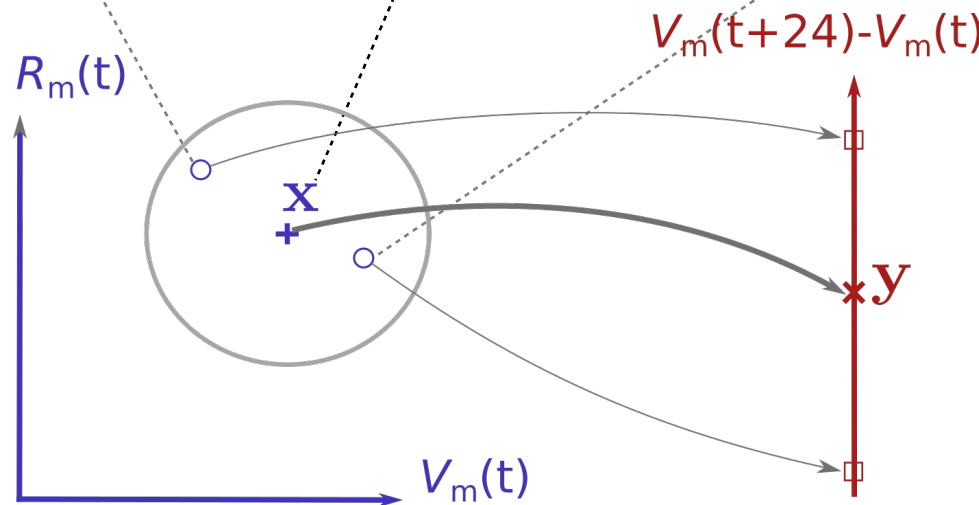
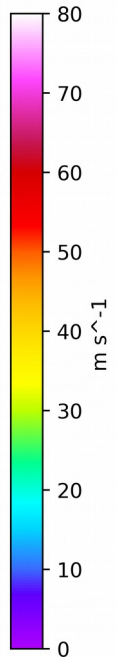
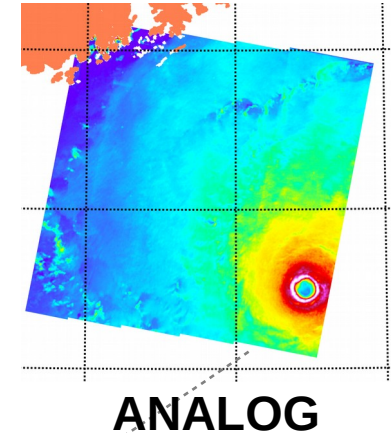
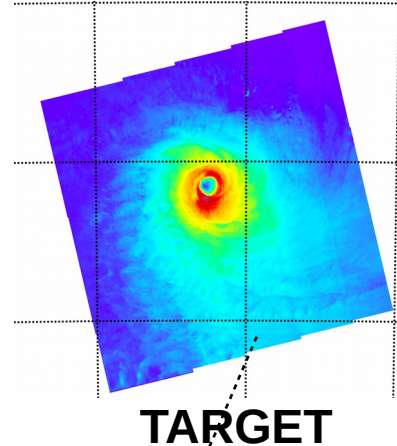
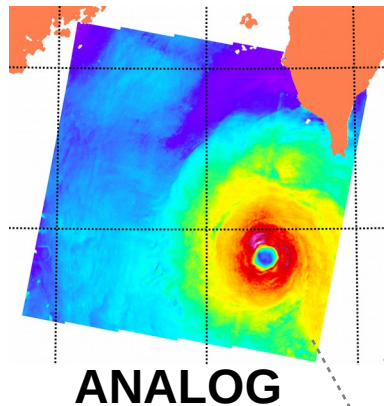
# Analog methods

- Applications : **Tropical cyclone intensity forecasts**  
(illustrative Synthetic Aperture Radar images from <https://cyclobs.ifremer.fr/app/>)

Krathon 2024-09-30

Anggrek 2024-01-27

Yagi 2024-09-04

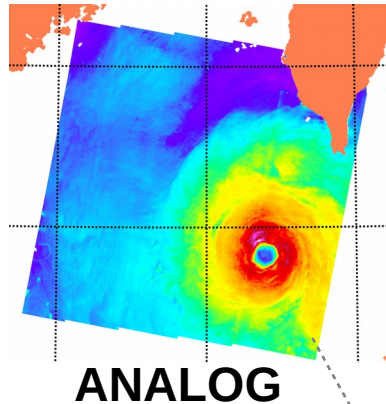




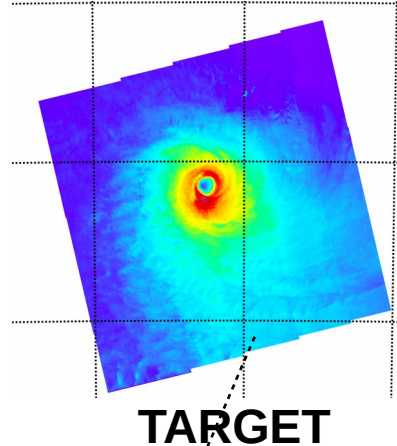
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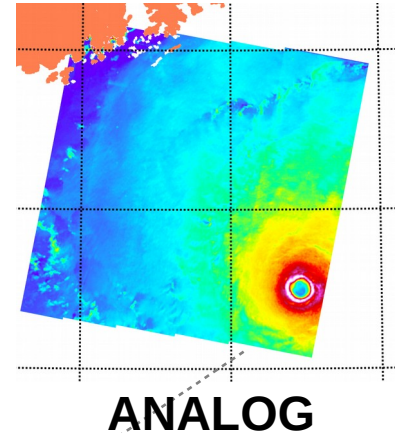
Krathon 2024-09-30



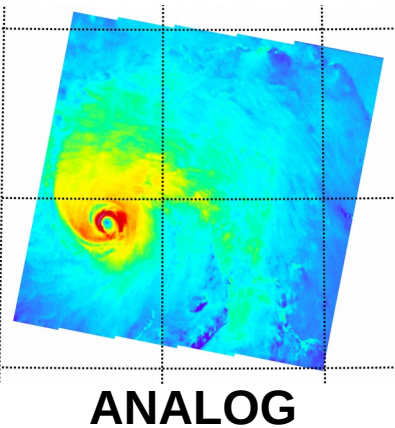
Anggrek 2024-01-27



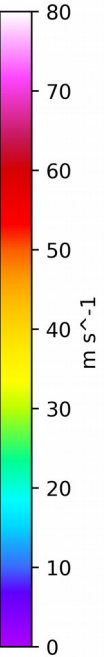
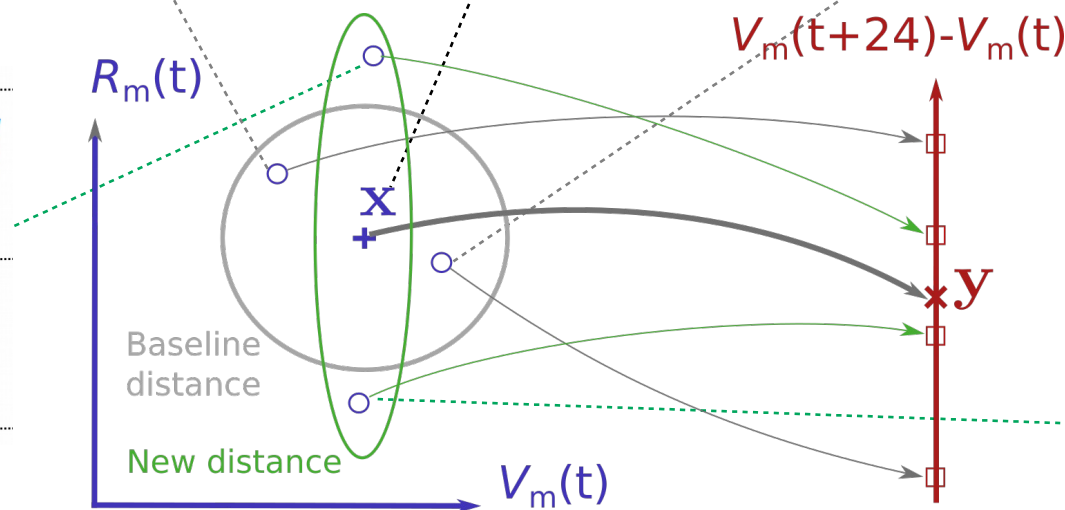
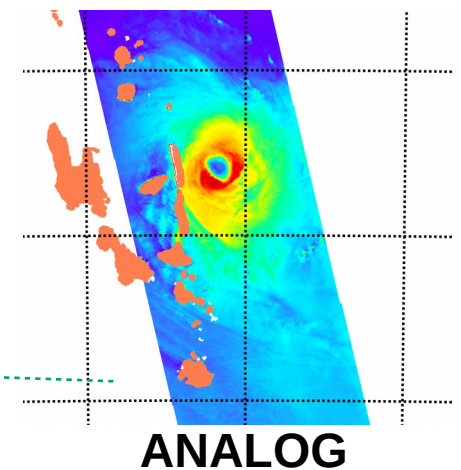
Yagi 2024-09-04



Lee 2023-09-08



Lola 2023-10-24



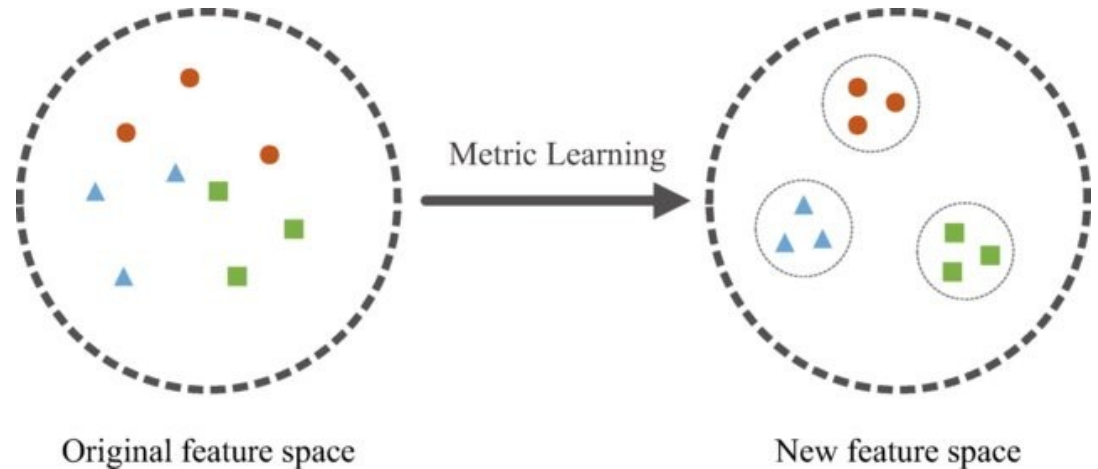
# Litterature review : optimal analog distance ?

- « **Try out** several distances » :
  - Toth (1991) → atmospheric circulation forecast
  - Matulla et al. (2008) → precipitation downscaling
- « Optimal **forecast** of dynamical systems » (KAF) :
  - Alexander and Giannakis (2020)
- « Weighting strategy **designed from scratch** » :
  - Fraedrich and Rückert (1998) → *general*
  - Alessandrini et al. (2018) → tropical cyclone ensemble reforecast

# Metric learning

- Pioneered by Xing et al. (2002)
- Idea :  
Use **gradient descent** (as in *machine learning*) to optimize *distance parameters* for a given *cost function*.

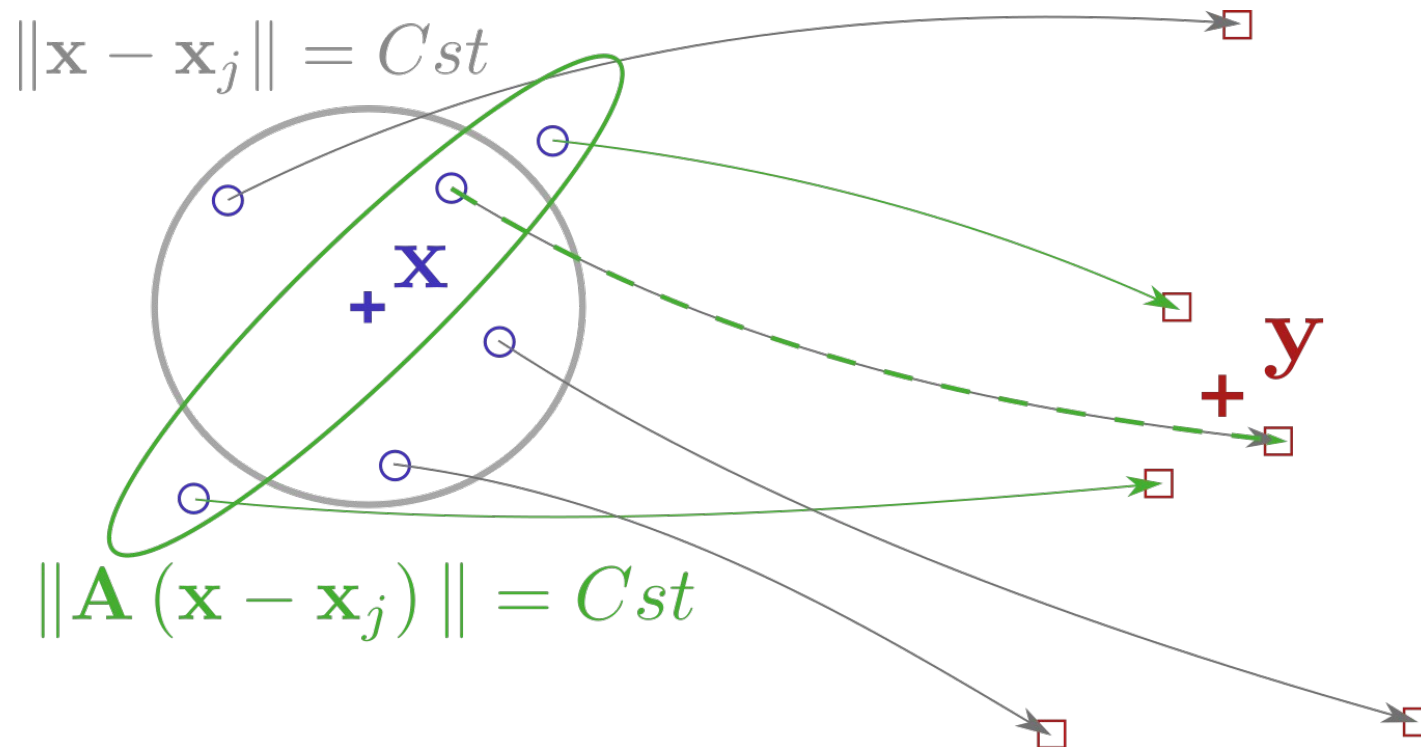
Figure from Qiao et al. (2022) :



- Example :  
Minimize a *forecast error* by adapting the weights in an analogy criterion.
  - Never used before in analog methods for atmosphere & ocean
  - **Our aim = bridge the gap between metric learning and analog methods**

# Idea of our algorithm

- Find linear transformation that minimizes a cost function (Continuous Ranked Probability Score, Mean-Squared Error)



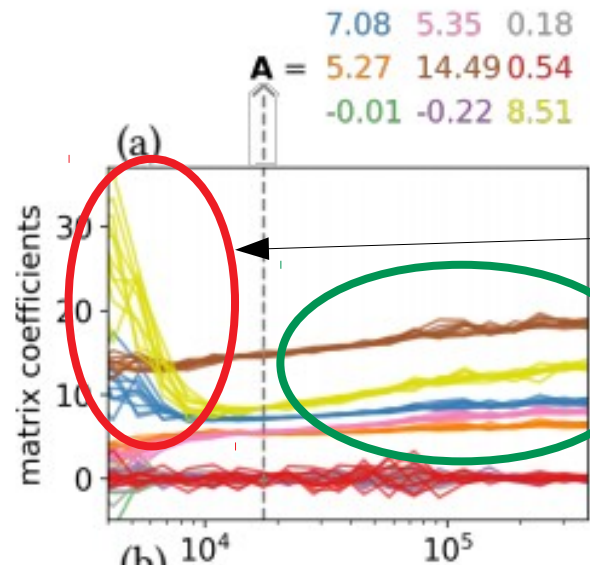


# Results : Idealized System (Lorenz 1963)

Test on idealized chaotic, 3-variable system (Lorenz 1963) allows to **vary the amount of available data**.

Lessons learned :

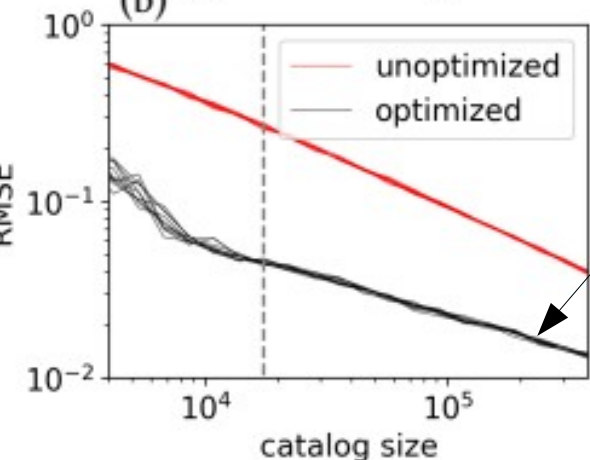
« parameters »



1) Small-size dataset  
→ *selective choice of analogs*.

2) The optimal distance *stabilizes for large-size datasets*.

« error »



3) Our algorithm allows to *reduce the amount of data needed to improve analog forecasting performances*  
→ relevant for small-size datasets (e.g. tropical cyclones datasets).

« amount of data »

# Results : Tropical Cyclones

Dataset: IBTrACS(2003-2022), North-Atlantic

Conditions:  $V_{\max} > 17\text{m/s}$ ,  $\max_t(V_{\max}) > 33\text{m/s}$

→ 110 cyclones

Objective: Weight 15 input variables for **analog intensity forecast**

$$\mathbf{x} := \left\{ V_{\max}(t), R_{\max}^{IBT}(t), R_{34}(t), f_{Cor}(t), u_{trans}(t), \right.$$

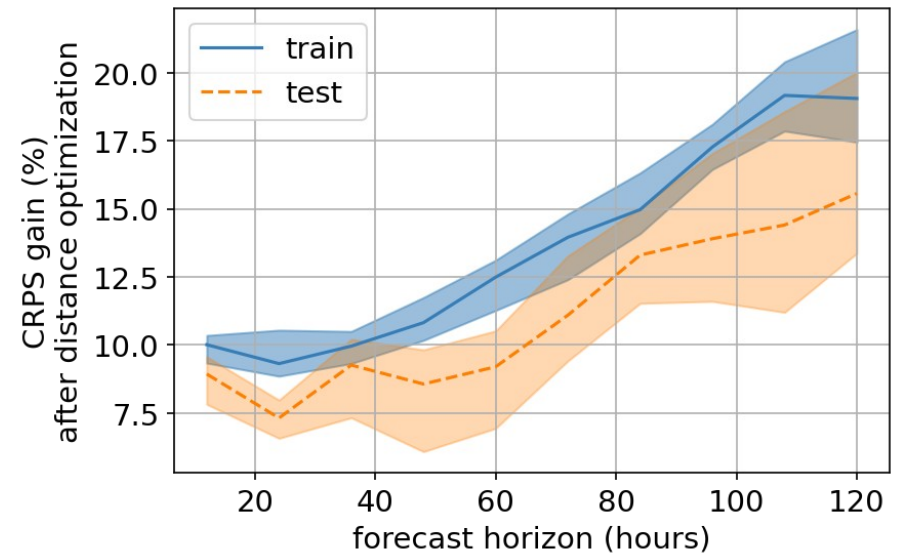
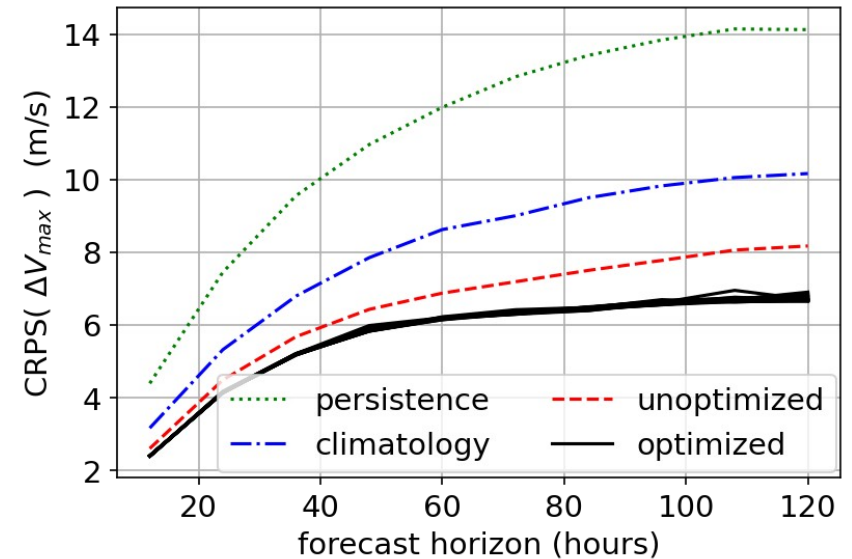
$$v_{trans}(t), R_{\max}^{A23}(t), \frac{dV_{\max}}{dt}(t), \frac{dR_{\max}^{IBT}}{dt}(t),$$

$$\frac{dR_{34}}{dt}(t), \frac{df_{Cor}}{dt}(t), \frac{du_{trans}}{dt}(t),$$

$$\left. \frac{dv_{trans}}{dt}(t), \frac{dR_{\max}^{A23}}{dt}(t), T_{18}(t) \right\}.$$

$$y := \Delta V_{\max}(t, h) := V_{\max}(t + h) - V_{\max}(t)$$

→ *Better than benchmarks and comparable to reference statistical forecasts (SHIPS)*



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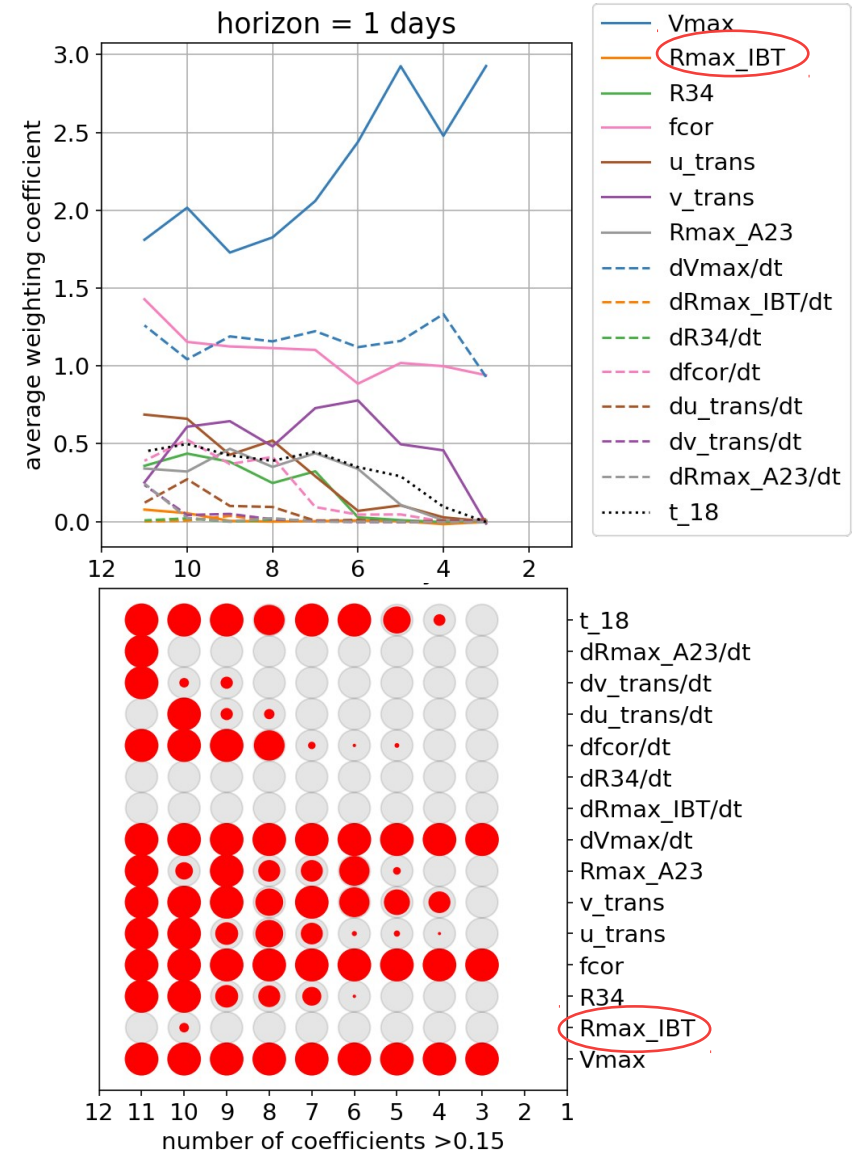
$$\frac{dR_{34}}{dt}(t), \frac{df_{Cor}}{dt}(t), \frac{du_{trans}}{dt}(t),$$

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$$y := \Delta V_{max}(t, h) := V_{max}(t + h) - V_{max}(t)$$

→ Selection of relevant variables

→ Confirmation of problems with  $R_{max}^{IBT}(t)$



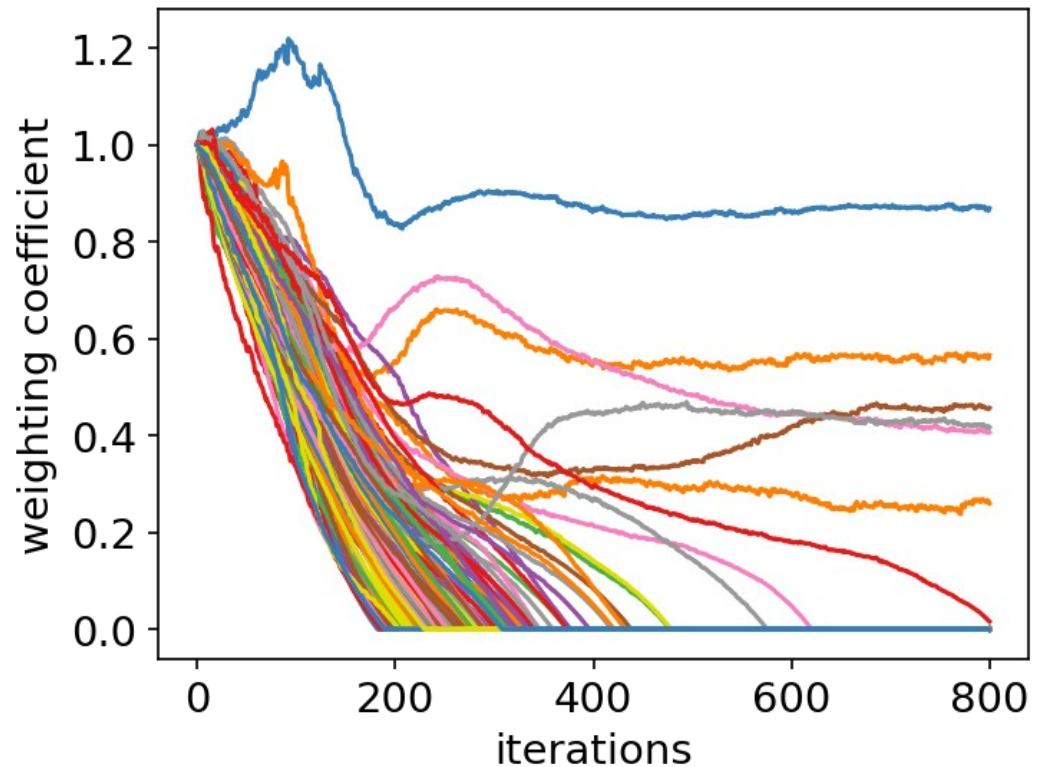
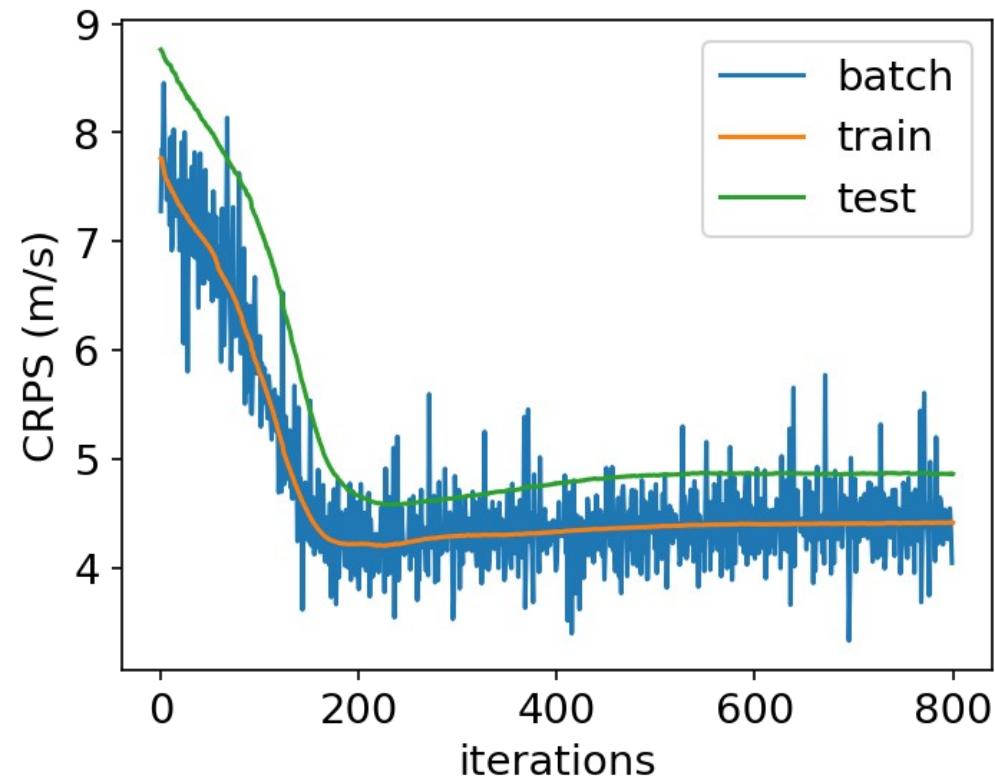
# Analog intensity forecast from SHIPS data variables

Test with **118 input variables** from SHIPS dataset.

Dataset = 769 tropical cyclones (2615 time-steps, 6h resolution) in East-Pacific (1982-2023)

→ Parcimonious distance (6 vars.):

|                |             |             |
|----------------|-------------|-------------|
| VMAX : 0.86    | INCV : 0.56 | VMPI : 0.45 |
| XD26 : 0.41    | SHGC : 0.40 | NAGE : 0.26 |
| Others : < 0.1 |             |             |



# Conclusion

- The distance is a key element of analog methods
- Machine learning algorithms exist to optimize it
- We adapted algorithms for « weather score » CRPS
- Results on L63: depend on #data (and objective)
- Result on TC :
  - suited for real-life datasets
  - allows to extract relevant features from IBTrACS and SHIPS datasets



# Perspectives

- Added-value of Synthetic Aperture Radar (SAR) data ?
- Analog-based prediction of rapid intensification

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→ to download Synthetic Aperture Radar data visit :  
<https://cyclobs.ifremer.fr/app/> (also see demo animation at <https://odl.bzh/NcEKUOtc>)
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# TC-related projects at LOPS-SIAM

- **OROWSHI** (Offshore wind turbine design: including joint Wind-wave information in Standard for Hurricane-exposed sites, ANR-FEM)
  - Améliorer la représentation statistique des paramètres des cyclones
  - Créer des modèles paramétriques des vents de surface et à hauteur des pales plus réalistes à partir d'observations et modélisations haute résolution
  - Evaluer l'effet du relief terrestre sur les fermes éoliennes offshore
- **CYCLOBS** <https://cyclobs.ifremer.fr/app/>
  - Développer des algorithmes et traitements appropriés à l'évaluation des vents extrêmes des cyclones tropicaux à partir de mesures SAR et radiomètre en bande L
  - Proposer un catalogue unique de mesures satellite dans les cyclones
- **CYMS** - Cyclone Monitoring Service based on Sentinel-1 <https://www.esa-cyms.org>
  - CYMS provides with SAR ocean surface wind speed data derived from Sentinel-1 along Tropical Cyclone tracks in near-real time and archived mode.
- **COWS** (CFOSAT for Ocean Waves Study)
  - Analyse consistante des vents et vagues au sein des cyclones
  - Evolution de la houle et estimation d'atténuation
  - variabilité à fine échelle et variabilité des vagues
- **MAXSS** (Marine Atmosphere eXtreme Satellite Synergy)
  - The primary objective of the ESA Marine Atmosphere eXtreme Satellite Synergy (MAXSS) project is to provide guidance and innovative methodologies to maximize the synergetic use of available Earth Observation data (satellite, in situ) to improve understanding about the multi-scale dynamical characteristics of extreme air-sea interaction.



# Thank you !